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GRAPHIC ORGANIZER ACTIVITY

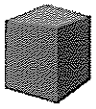
Name: .....

Date: .....

SCIENCE > MATTER AND CHEMISTRY > STATES OF MATTER

MATRIX

For any given substance, describe how each of the following properties is different from state to state. If a property does not change from one state to the next, note that on the matrix.

	 SOLID	 LIQUID	 GAS
MASS	....	....	....
TEMPERATURE	....	....	....
VOLUME	....	....	....
CHEMICAL STRUCTURE	....	....	....
PARTICLE MOVEMENT	....	....	....

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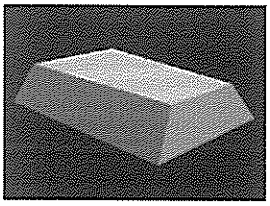
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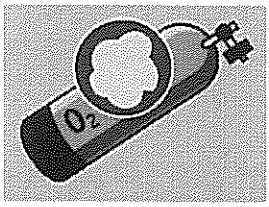
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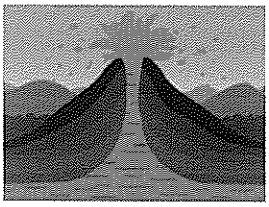
SCIENCE > MATTER AND CHEMISTRY > STATES OF MATTER

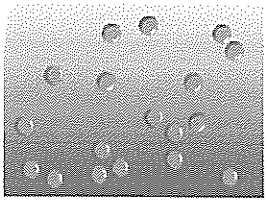
MATCHING

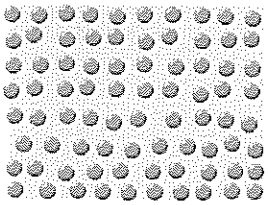
Match the object on the left with the configuration of its molecules on the right.

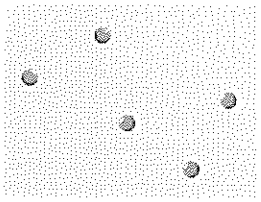












TRUE OR FALSE

Determine if the statement is true or false. If false, replace the word(s) in bold with the correct word(s) on the line. If true, write 'true' on the line.

1. ..... All matter is made up of tiny particles called **ATOMS**.
2. ..... There is a **CHEMICAL** difference between the gaseous and liquid state of the same substance.
3. ..... Liquids have a fixed **TEMPERATURE**, but no fixed shape.
4. ..... Molecules of ice have less **MASS** than molecules of water.
5. ..... A **PLASMA** is a gas with an electrical charge.

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VOCABULARY ACTIVITY

Name: .....

Date: .....

SCIENCE > MATTER AND CHEMISTRY > STATES OF MATTER

DEFINE Can you explain the terms below in your own words?

MATTER:

.....

ATOM:

.....

MOLECULE:

.....

LIQUID:

.....

RIGID:

.....

GAS:

.....

SOLID:

.....

PLASMA:

.....

COLLOID:

.....

ABSOLUTE ZERO:

.....

Choose an additional term from the movie to define.

.....

.....

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STATES OF MATTER

Date:_____

Name:_____

Class:_____

1 What is matter? Choose the best answer.

A Anything that is solid

B Anything that takes up space

C Anything that has a fixed volume

D Anything that can be seen

2  **In what way are liquids different from solids?**

A Liquid molecules are lighter than solid molecules

B Liquids are made of atoms; solids are made of molecules

C Liquids don't have a fixed shape; solids do

D Liquids are always denser than solids

3 What happens to the chemical structure of water when it changes state?

A Water molecules break apart to form individual atoms

B Water molecules harden into ice molecules

C Water molecules melt into gas molecules

D Nothing happens to the chemical structure

4 What will happen if you keep increasing both the pressure and temperature of a liquid?

A It will boil

B It will transpire

C It will melt

D It will solidify

5 What do lightning and stars have in common?

A Both have no electrical charge

B Both contain plasma

C Both have a fixed volume

D Both contain all three states of matter

6 Water's chemical formula is "H2O." What does this mean?

A Water is composed of hydrogen molecules and oxygen molecules

B A gram of water contains 20 hydrogen atoms

C Each water molecule contains two hydrogen atoms and one oxygen atom

D Water molecules cannot be split into smaller pieces

7 What is the basic shape of a liquid?

A A sphere

B The same shape as a gas, only rounder

C A cube

D Whatever the shape of its container is

8 How can you remove energy from matter?

A By increasing its volume

B By lowering its temperature

C By increasing its pressure

D By boiling it

9 Gases have a tendency to expand. What's the best synonym for "expand?"

A Collide

B Spread out

C Shrink

D Boil

10 Where would you find a Bose-Einstein condensate?

A Inside stars

B At the North Pole

C In a science lab

D Bose-Einstein condensates exist only in theory

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BrainPOP ACTIVITY

Name: .....

Date: .....

SCIENCE > MATTER AND CHEMISTRY > MATTER CHANGING STATES

WORD SCRAMBLE

Unscramble the words to complete each sentence.

1. **MRATET** is anything that has mass and takes up space.

   .....

2. The three basic states of matter are **ILSOD**, **UDLIIQ**, and **ASG**.

   ....

3. A state change is a **PCLYHASI** change.

   .....

4. Matter changes state because of **UEREPRSS** and **RTEEAUEMTRP**; **LECMCIHA** properties do not change.

   ....

5. Bonded water molecules form a crystal **TIACTLE** structure.

   .....

6. **EGAINHT** gives molecules energy that causes them to vibrate faster, breaking their bonds and melting.

   .....

7. The temperature at which a solid melts into a liquid is its **LMITEGN** point.

   .....

8. Heat energy that helps break the molecular bonds of a solid is heat of **FOSUNI**.

   .....

9. Gas molecules have more energy than liquid molecules and bounce around **RYADONML**.

   .....

10. Heat energy required to break the attraction of liquid molecules is heat of **TPAAIOIORZVN**.

    .....

11. The temperature at which a liquid turns into a gas is the **OBNGLII** point.

    .....

12. **TMOILASINBU** is when a solid turns straight into a gas.

    .....

FURTHER RESEARCH

What is the process of a gas changing back into a liquid called?

.....

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BrainPOP GRAPHIC ORGANIZER ACTIVITY

Name: .....

Date: .....

SCIENCE > MATTER AND CHEMISTRY > MATTER CHANGING STATES

FLOW CHART

Write the steps that describe the process of water changing from a solid to a liquid and from a liquid to a gas.

ICE

.....

.....

.....

.....

.....

LIQUID WATER

.....

.....

.....

.....

.....

STEAM

.....

.....

.....

.....

.....

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ACTIVITY VOCABULARY

Name: .....

Date: .....

SCIENCE > MATTER AND CHEMISTRY > MATTER CHANGING STATES

DEFINE Can you explain the terms below in your own words?

MATTER:

.....

MASS:

.....

VAPOR:

.....

MOLECULE:

.....

LATTICE:

.....

MELTING POINT:

.....

BOND (IN CHEMISTRY):

.....

FUSE (VERB):

.....

RANDOM:

.....

SUBLIMATION:

.....

Choose an additional term from the movie to define.

.....

.....

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MATTER CHANGING STATES

Date:_____

Name:_____

Class:_____

1 What are the three main states of matter?

A Solid, liquid, and gas

B Ice, water, and steam

C Temperature, pressure, and energy

D Hot, cold, and warm

2 Compared to ice, how do molecules of water behave?

A They move less freely

B They are locked into a crystal lattice

C They move more freely

D They bounce off one another randomly

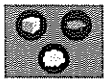
3 What is matter?

A Anything that is visible

B Anything that has mass and takes up space

C Anything that reflects light

D Anything that absorbs light

4  **What do water vapor, liquid water, and ice have in common?**

A The same crystal lattice structure

B The same amount of energy

C The same physical properties

D The same chemical properties

5 What's the easiest way to add energy to matter?

A Heating it

B Cooling it

C Decreasing pressure

D Decreasing temperature

6 The melting point of H2O is 0 degrees Celsius. This is the same as its:

A Boiling point

B Vaporization point

C Fusion point

D Freezing point

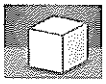
7 What is it called when a solid turns directly into a gas?

A Sublimation

B Vaporization

C Melting

D Fusion

8  **Dry ice has the same chemical properties as what gas?**

A Helium

B Hydrogen

C Nitrogen

D Carbon dioxide

9 How do the molecules of a gas behave?

A They vibrate in place

B They bounce around randomly

C They are locked in a crystal lattice

D They clump together

10  **If you put a thermometer into a pot of melting ice, when will the temperature rise past zero?**

A As soon as the ice begins to melt

B After most of the ice has melted

C When the ice begins turning into steam

D When all the ice has melted

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VOCABULARY ACTIVITY

Name: .....

Date: .....

SCIENCE > MATTER AND CHEMISTRY > PROPERTY CHANGES

DEFINE Can you explain the terms below in your own words?

PHYSICAL CHANGE:

.....

.....

.....

STATE CHANGE:

.....

.....

.....

PHYSICAL WEATHERING:

.....

.....

.....

CHEMICAL CHANGE:

.....

.....

.....

Choose an additional term from the movie to define.

.....

.....

.....

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PROPERTY CHANGES

Date:_____

Name:_____

Class:_____

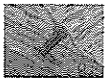
1 Which of the following is an example of a physical change?

A Water freezing into ice

B A piece of wood burning

C A toy car rusting

D Zinc producing hydrogen gas when mixed with water

2  **What type of change has occurred when a nail rusts?**

A A physical change

B A chemical change

C An atomic change

D A reactive change

3 What does it mean when there is a physical change?

A Matter has changed from one substance into another

B Matter has changed on the molecular level

C A new substance has been formed

D Matter has changed size, shape or form

4 What is an example of a chemical change that happens inside your body?

A Food being broken down by enzymes in your stomach

B Food being broken into small pieces by your teeth

C Food being moved from your esophagus into your stomach

D Water being absorbed inside your large intestine

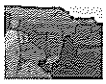
5 What is an example of a physical change that happens inside your body?

A Food being broken down by enzymes in your stomach

B Food being broken into small pieces by your teeth

C Food being moved from your esophagus into your stomach

D Water being absorbed inside your large intestine

6  **What can you infer about the changes that formed the Grand Canyon?**

A They were molecular changes

B They were chemical changes

C They were physical changes

D They were changes in weather

7 How does rust form?

A Through paint flaking off of a metal surface

B Through a chemical reaction between iron and oxygen

C Through weathering

D Through the physical mixing of iron and water

8 What happens when sodium undergoes a chemical reaction with chlorine?

A The two substances form a heterogeneous mixture

B The two substances form a homogenous mixture

C A new compound is formed – salt

D The two substances undergo various physical changes

9 Which term best describes the two substances that react to form salt?

A Delicious

B Nutritious

C Edible

D Toxic

10 Which phrase most accurately describes a chemical change?

A A change on the molecular level

B A change in appearance

C A change in taste

D A change in form

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